



Sultan Mahmud Rana



[gsmrana.github.io](https://github.com/gsmrana)



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Dhaka, Bangladesh

Nationality: Bangladeshi

Professional Summary

Principal Embedded Software Engineer with 12+ years of experience in firmware architecture, device drivers, real-time systems, and wireless communication technologies. Designing and optimizing firmware for automotive, wireless sensor networks, IoT, GPS tracking, cold chain monitoring and medical devices. Expert in real-time systems, low-power optimization, and safety-critical software. Strong background in C/C++, RTOS, ARM Cortex-M, with proven ability to lead teams, mentor engineers, and deliver high-reliability products from concept to market.

Key Technical Skills

- ✚ **Programming:** C, C++, Python, Embedded Assembly (Bare-metal firmware)
- ✚ **MCUs & Platforms:** ARM Cortex-M, STM32, Atmel, Microchip, PIC, OKI, Renesas, Lapis, TI
- ✚ **Cellular Network:** GSM, GPRS, 2G, 3G, 4G, LTE-M, CAT-M
- ✚ **Real-Time Location:** GPS, GNSS, Cellular Triangulation
- ✚ **IoT Memory:** EEPROM, Flash Memory, USB Drive, PDF generation
- ✚ **RF Protocols:** BLE, WiFi, Zigbee, 815/915MHz and 2.4GHz ISM band
- ✚ **RTOS/OS:** FreeRTOS, Embedded Windows CE 5.0, Embedded Linux
- ✚ **Bootloader:** USB, Serial and OTA firmware upgrade.
- ✚ **Communication Protocols:** UART, I²C, SPI, CAN, USB CDC/MSC, MQTT, TCP/IP, Socket communication, Protobuf and Base64 encoding/decoding.
- ✚ **IDE/Frameworks:** IAR Embedded Workbench, Microchip Studio, Atmel Software Framework (ASF), STM32 Hardware Abstraction Layer (HAL), Keil, Renesas CS+ for CC, OKI U8
- ✚ **Debugging Tools:** JTAG, SW, ISP, Oscilloscope, Logic Analyzer, Power Analyzer
- ✚ **DevOps & Tools:** GCC, GDB, Git, SVN, SourceTree, GitLab CI/CD, Docker (cross-compilation)
- ✚ **Linux Kernel:** Buildroot, Yocto Project, Raspberry Pi, BeagleBone Black
- ✚ **Safety Standards:** MISRA-C, ISO 26262 (ASIL-B), IEC 62304, DO-178C

Professional Experience

ReliSource Technologies Ltd. – Dhaka, Bangladesh

Headquartered: Bedford, MA, USA | www.reli-source.com

Principal Embedded Software Engineer | Sep 2022 – Present

- Served as Technical Lead, guiding the embedded team through architectural decisions, providing technical direction, conducting code reviews, and mentoring engineers to ensure high-quality deliverables.
- Designing and Developed file system corruption protection system for a USB MSC device.

- Designed firmware architecture from scratch for a temperature and humidity data loggers and location tracker device.
- Implemented device drivers for GSM/LTE cellular modems, GPS/GNSS real Time and Location and NDIR CO₂ sensors.
- Designed and deployed bootloaders for Microchip SAM4S controller for firmware upgrade over USB and OTA and to support FTM testing.
- Integrated USB CDC driver into an Embedded Windows CE 5.0 handheld device that used as a portable data collector for non-real-time data logger.

Senior Embedded Software Engineer | Dec 2018 – Sep 2022

- Implemented sensor driver for various temperature and humidity monitoring device that used in cold chain monitoring industry.
- Implemented low-level drivers for SPI, I²C, UART across multiple MCU families.
- Enhanced feature of a wireless sensor network, wireless repeaters, and gateway system solutions.

Embedded Software Engineer | Oct 2015 – Dec 2018

- Developed new feature and bug fixes of a preoperatory protocol RF network and node control devices.
- Enhanced drivers' firmware of 915 MHz and 2.4GHz Radio chips for RF communication.
- Developed a Desktop tool and supporting firmware for analyzing packets of a wireless sensors large-scale deployment.

Pi Labs Bangladesh Ltd. - Dhaka, Bangladesh

Product Development and Research Engineer | May 2014 – Oct 2015

- Developed firmware for a vehicle tracking system (Prohori) device which is designed and manufactured completely in Bangladesh.
- Investigated the feasibility of integrating car's OBD port with the VTS device for monitoring card diagnosis data remotely.
- Designing and developed a digital weight machine hardware and firmware which supports reporting weights to a server. (Used in Rokomari.com)

Open-Source Projects & Contributions

- Design and developed a Table Tennis scoreboard hardware and firmware from scratch using dot matrix display and RF communication.
- Designed a Serial COM desktop tool which is very useful for developing firmware for UART or USB CDC or Serial communication.
- Designed a tool for USB MSC disk drive analyzer desktop tool which is very used for analyzing and developing USB Mass Storage device firmware.

Education

- M.Sc. Computer Science and Engineering – United International University (UIU) Dhaka, Bangladesh, 2025
- B.Sc. Computer Science and Engineering – Canadian University of Bangladesh (CUB) Dhaka, Bangladesh, 2023
- B.Sc. Mechanical Engineering – Khulna University of Engineering & Technology (KUET) Khulna, Bangladesh, 2014

Publications

- MedBot: An IoT Based Real-Time Healthcare Monitoring and Telemedicine Solution System, *CIIR 2023, India (Springer)*.
- Design and Implementation of a Remote-Controlled Robotic Arm Based on Industrial Application Perspective. *ICMIME 2013, RUET*.
- Design and Implementation of a GSM Based remote home security and appliance control system. *ICAEE 2013, IUB (IEEE Xplore Online Digital Library)*.
- Design & Implementation of a Robot for Maze-Solving with turning indicators using PID controller. *ICIEV 2013, DU (IEEE Xplore Online Digital Library)*.
- Development of a Remote-Control System for an Unmanned Vehicle Using Mobile Phone Based Network. *ICMERE 2013, CUET*.
- Implementation of Drive by Wire technology replacing the conventional vehicle control system. *ICMIME 2013, RUET*.

Awards

- Secured 1st runner up prize at International Autonomous Robotic Competition, *IARC 2013, Dhaka, Bangladesh*.
- Secured Champion prize in Poster Session at Int. Conf. on Information & Communication Tech. *EICT 2014. Khulna, Bangladesh*.

Patent

- Universal Smart Energy Meter with Automatic Load Management (New Invention) *Department of Patents, Design & Trademark, Ministry of Industry, Bangladesh 2015*.